

# IAM Password Recovery Performance Report

## Overview

This document captures the performance characteristics of the password recovery workflow within the IAM system.

**Environment:** Local Development

**Base URL:** http://localhost:8080

**Date:** 18 June 2026

**Scope:** Password Recovery APIs (Successful Requests Only)

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## 1. Forgot Password

### Password Reset Request

| Endpoint                     | Method | Status | Response Time |
|------------------------------|--------|--------|---------------|
| /api/v1/auth/forgot-password | POST   | 200    | 7364 ms       |

### Observation

- Slowest endpoint observed in the authentication module.
- Endpoint performs multiple operations including:
  - User validation
  - Reset token generation
  - Token persistence
  - Email preparation and delivery
  - Audit logging

### Optimization Considerations

- Measure email delivery time separately.
  - Profile token generation and database operations.
  - Consider asynchronous email dispatch.
  - Analyze audit logging overhead.
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## 2. Reset Password

### Password Reset

| Endpoint                    | Method | Status | Response Time |
|-----------------------------|--------|--------|---------------|
| /api/v1/auth/reset-password | POST   | 200    | 984 ms        |

### Observation

- Response time remains under one second.
- Performance is acceptable for a security-sensitive operation.
- Endpoint likely includes:
  - Reset token validation
  - Password hashing
  - Password update
  - Audit log creation

### Optimization Considerations

- Measure password hashing duration.
- Profile database update operations.
- Verify audit logging impact.

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## Summary Metrics

| Metric                | Value   |
|-----------------------|---------|
| Fastest Endpoint      | 984 ms  |
| Slowest Endpoint      | 7364 ms |
| Average Response Time | 4174 ms |
| Success Rate          | 100%    |

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## Key Findings

### 1. Forgot Password is the Primary Bottleneck

The forgot-password workflow exhibits significantly higher latency compared to all other authentication endpoints tested so far.

## 2. Reset Password Performance is Acceptable

The password reset operation completes in under one second while performing security-critical operations.

## 3. Email Processing Requires Investigation

Given the large difference between forgot-password and reset-password execution times, email generation and delivery are likely contributors to overall latency.

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## Optimization Priority

### High Priority

- Forgot password workflow latency
- Email dispatch performance
- Token persistence operations

### Medium Priority

- Password reset execution time
- Audit logging overhead

### Low Priority

- Minor response time improvements after bottlenecks are identified
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## Notes

- Metrics were collected from a local development environment.
- Values represent baseline measurements prior to optimization efforts.
- This document will be used for future performance comparison and benchmarking.